Document 8. Market Failure: Merit / Demerit / Public Goods + Info Failure

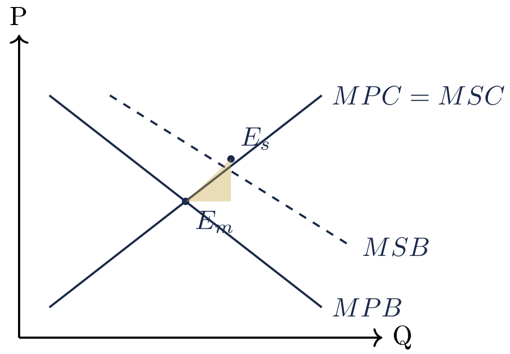
Topic overview. Beyond externalities, the market also fails when goods are mis-valued by consumers (merit / demerit), when goods are non-rival and non-excludable (public goods), or when information is imperfect.

Section A. Important Definitions

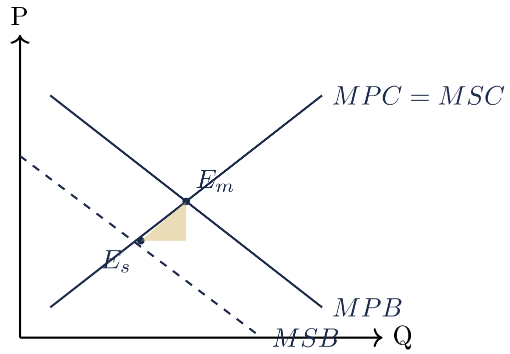
Merit good	A good that consumers under-value due to imperfect information about long-run private benefits (e.g. healthcare, education, retirement savings).
Demerit good	A good that consumers over-value because they under-estimate private costs (e.g. tobacco, alcohol, gambling).
Public good	A good that is *non-rival* (consumption by one does not reduce availability) and *non-excludable* (cannot prevent free riders). e.g. national defence, lighthouses, street lighting.
Free-rider problem	Inability of producers to charge for a non-excludable good; private firms have no incentive to supply it.
Quasi-public good	Partly rival or partly excludable (e.g. congested public road, broadcast TV).
Asymmetric information	One party has more information than the other in a transaction.
Adverse selection	Pre-contract information asymmetry: the wrong types of buyers / sellers enter the market (lemons problem).
Moral hazard	Post-contract information asymmetry: parties change behaviour once insured / protected.

Section B. Diagram — Merit / Demerit Goods (mis-valuation)

Merit good (under-consumed)



Demerit good (over-consumed)



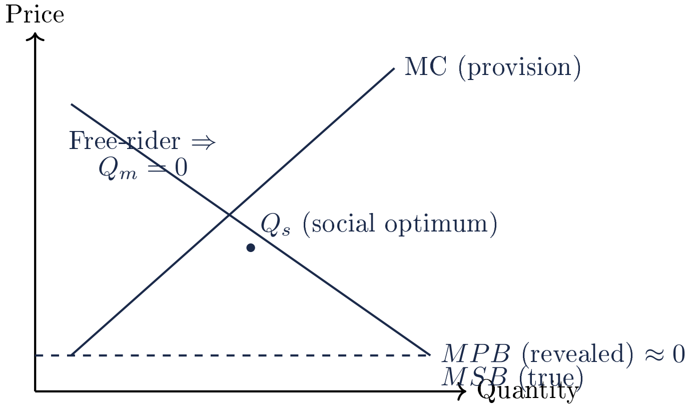
Section C. Question Type 1: Explain why a merit / demerit good causes market failure

Step 1. Define merit / demerit good; identify the information failure cause.
Step 2. Free market produces at $MPB = MPC$ giving Q_m.
Step 3. If we add information correction, true valuation gives MSB curve. Social optimum at $MSB = MSC$ giving Q_s.
Step 4. For a merit good, $Q_m < Q_s \Rightarrow$ under-consumption; for a demerit good, $Q_m > Q_s \Rightarrow$ over-consumption.

Step 5. Welfare loss = DWL triangle bounded by MSB, MSC, and the relevant axis.
Step 6. Conclude that the market fails because consumers do not correctly perceive the full private benefits / costs.

NOTE: Some teachers prefer to combine merit goods with positive externalities. The 9570 syllabus treats them as separate causes: merit good is internal *mis-valuation; externality is* external *effect on third parties. Clearly identify which source of failure your question references.*

Section D1. Diagram — Public Good Missing Market



Caption. Because the good is non-excludable, consumers free-ride and reveal $MPB \approx 0$. No private firm has incentive to produce. The socially optimal quantity Q_s requires government direct provision financed by taxation.

Section D. Question Type 2: Public Goods — explain why they would not be provided by the free market

Step 1. Define non-rivalry: one person's consumption does not reduce availability for others.
Step 2. Define non-excludability: cannot prevent free riders from consuming once provided.
Step 3. Hence private producers cannot charge a price; the free-rider problem means demand is under-revealed.
Step 4. Profit-maximising firms have no incentive to supply ⇒ **missing market**.
Step 5. Government must provide directly, funded by taxation. Examples: national defence, street lighting, public lighthouses, weather forecasting.

Section E. Question Type 3: Information Failure

Step 1. Define information failure: imperfect / asymmetric information leads consumers or firms to mis-allocate resources.
Step 2. Identify the type:

- *Imperfect information*: consumers mis-value private benefits or costs (merit / demerit goods).
- *Asymmetric information (adverse selection)*: e.g. used-car "lemons" market; insurance markets for high-risk individuals.
- *Asymmetric information (moral hazard)*: e.g. insured drivers driving more recklessly.

Step 3. Show the welfare consequence: under-consumption, over-consumption, or market collapse.
Step 4. Discuss remedies: education campaigns, compulsory disclosure, signalling (warranties, qualifications), screening (medical tests, deductibles).

Section F. Question Type 4: Evaluate government interventions

Source of failure	Best-fit policy	Evaluation
Merit good	Subsidy, direct provision, education campaign, mandate (compulsory education).	Direct provision crowds out private supply; subsidy uplifts producer surplus if PED is low.
Demerit good	Tax, regulation, ban, education, age limits.	Tax is regressive; ban creates black markets; education slow.
Public good	Direct provision financed by taxation.	Preference revelation problem; political-economy distortions.
Asymmetric info	Compulsory disclosure, accreditation, signalling, screening, regulation.	Compliance cost; capture by regulated industries.

Section G. Singapore Applications

- **Merit good — education**: MOE compulsory schooling, ITE / polytechnic / university subsidies; SkillsFuture credits.
- **Merit good — healthcare**: MediSave, MediShield Life, CareShield Life, subsidised polyclinics.
- **Demerit good — tobacco / alcohol**: excise duties, plain packaging, minimum legal age 21 (tobacco), Cap on alcohol after 10:30pm.
- **Public good — national defence**: SAF; **quasi-public — roads**: ERP introduces excludability via congestion pricing.
- **Asymmetric info — finance**: MAS regulates disclosure; CPF mandates retirement saving (paternalism + info failure).

Section H. Worked Examples

Worked Example 1: Healthcare subsidy

Q. Using a diagram, explain why healthcare is under-consumed and assess the government subsidy.

Healthcare is a merit good: consumers under-estimate long-run private benefits of preventive care. Free market at $MPB = MPC$ produces $Q_m < Q_s$. Subsidy = MEB shifts MPC down; new equilibrium at Q_s. *Limitations*: behavioural barriers; supply-side capacity; fiscal cost. Singapore complements subsidies with mandatory MediSave + targeted CHAS subsidies.

Worked Example 2: Tobacco tax

Q. Evaluate the use of taxation to reduce tobacco consumption.

Tax = MEC + perceived cost gap shifts MPC up. PED for tobacco is low so consumption falls modestly but revenue is high. Regressive on low-income smokers; black-market risk. Singapore combines tax with plain packaging, smoking-prohibited zones, and Health Promotion Board campaigns. Net effect: smoking prevalence has fallen to <10% of adults.